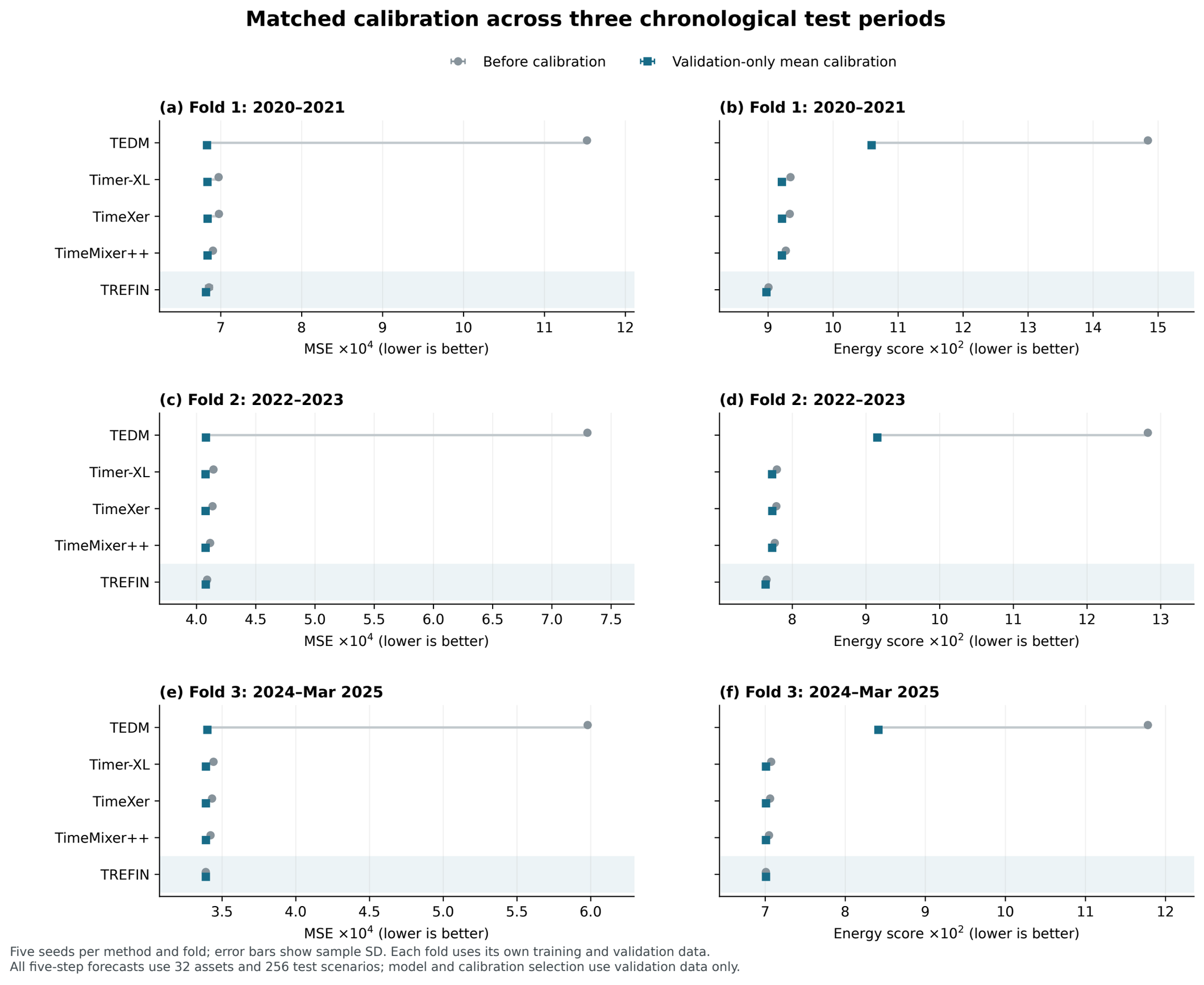
Matched calibration across three chronological test periods
Before calibration
Validation-only mean calibration
(a) Fold 1: 2020–2021
(b) Fold 1: 2020–2021
(c) Fold 2: 2022–2023
(d) Fold 2: 2022–2023
(e) Fold 3: 2024–Mar 2025
(f) Fold 3: 2024–Mar 2025
TEDM
Timer-XL
TimeXer
TimeMixer++
TREFIN
MSE ×10^4 (lower is better)
Energy score ×10^2 (lower is better)
Five seeds per method and fold; error bars show sample SD. Each fold uses its own training and validation data.
All five-step forecasts use 32 assets and 256 test scenarios; model and calibration selection use validation data only.